

Run 224460 Follow-up: Arms A/D Recovered, SiPM-Wall Channel Autopsy

n_TOF EAR2 X17 — run 224460 (July 16, after plastic operating-point change)

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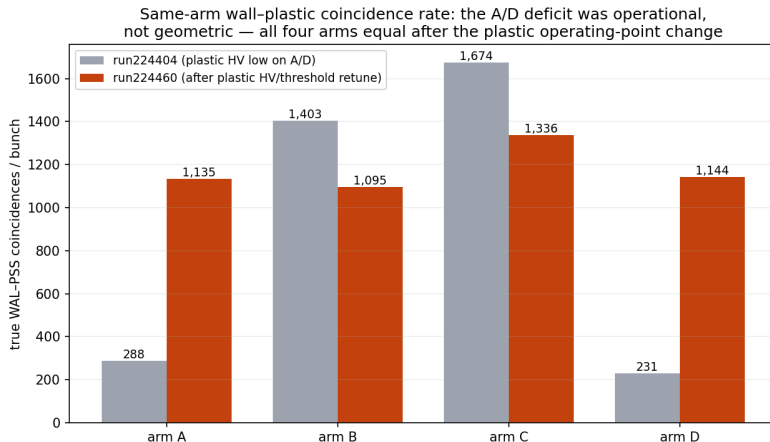
analysis and slides prepared with Claude (Anthropic AI assistant)

July 17, 2026

Data & selection

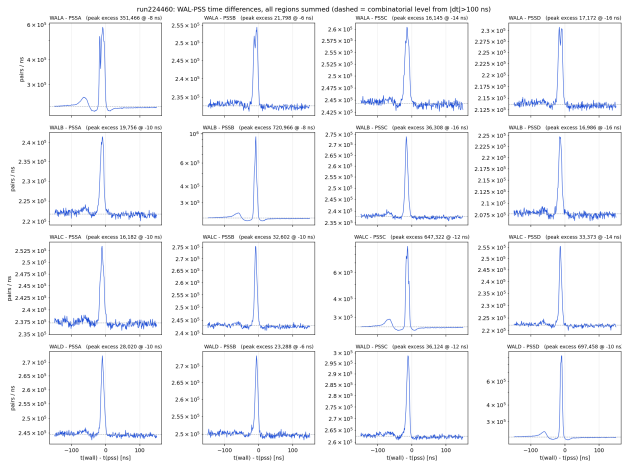
- ▶ Official processed file, run 224460: 4005 bunches (2058 dedicated), $\sim 335\text{M}$ WAL/PSS hits ($\sim 25\%$ more than 224404).
- ▶ **No SiPM-wall outage this run** — all 3943 beam-on bunches used (224404 lost 46%).
- ▶ **The official index tree is empty in this run** — pipeline now falls back to PKUP (one entry/bunch, identical intensities); the 224404 bunch mask is applied per run.
- ▶ Between the runs the **plastic PMT operating points changed**: every PSS channel's spectral structure moved up $\sim 2\times$; PSSA thresholds deepened ($-14/-13 \rightarrow -19/-18$ mV). Walls unchanged.
- ▶ Selection as before: beam-on bunches, hits > 0.1 ms after γ -flash.

Headline: the A/D coincidence deficit is gone

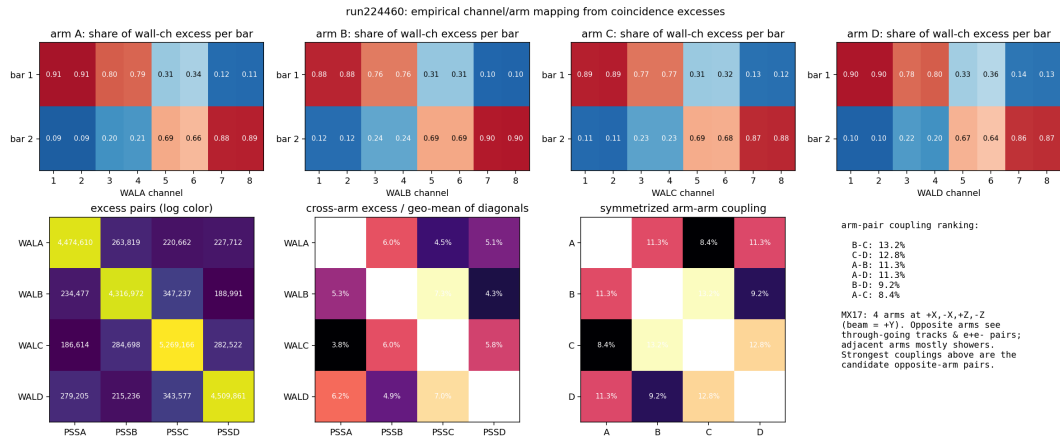


Same-arm true coincidences per bunch. A: 288 $\rightarrow$ 1135, D: 231 $\rightarrow$ 1144 (4–5 $\times$); B/C stable. All four arms now agree within $\pm 11\%$. **The 224404 deficit was the plastic operating point — not geometry, mapping, or cabling.**

All 16 wall-plastic pairs, calibrated dt



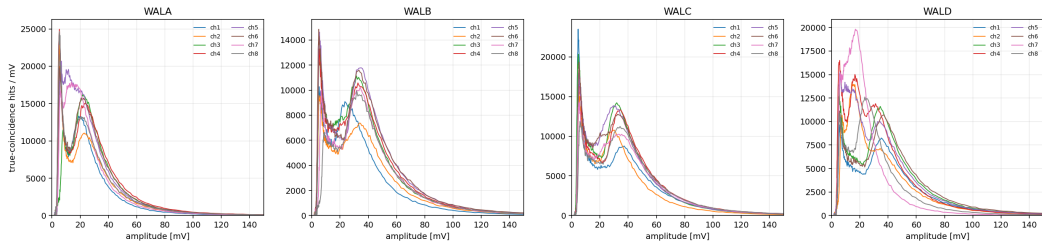
Channel mapping re-confirmed with full contrast



Bar-1 share falls 0.90→0.10 across channels 1→8 on every arm (A/D were washed out to 0.7/0.3 in 224404 by shower-dominated samples). Plastic ch1 = left, wall pairs (1,2)...(7,8) = groups left→right seen from the back — in stone.

MIP peaks on all 32 SiPM-wall channels

run224460: sideband-subtracted coincidence amplitude spectra (SiPM wall, tof > 0.1 ms, linear axes)

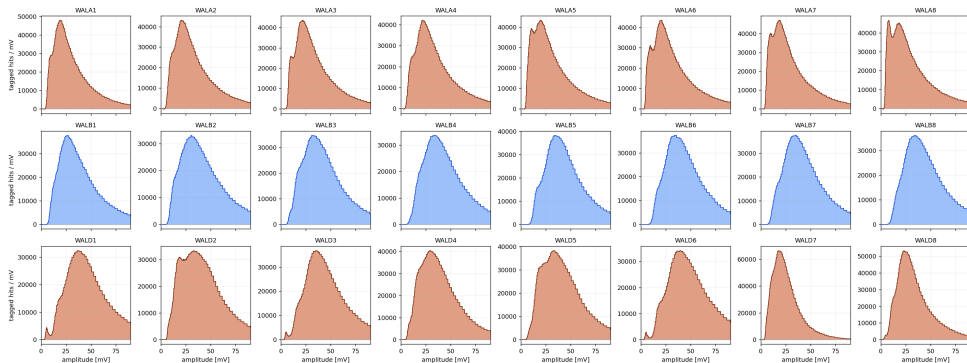


Sideband-subtracted coincidence spectra: two-component shape (accidentals ~ 5 mV + MIP) everywhere.

B/C/D₁₋₆: 33–43 mV. **WALA all channels $\sim 30\%$ low (22–28 mV); WALD ch7/ch8 low and distorted $\Rightarrow$ next slides.**

Odd-channel autopsy 1: partner-tagged response (no plastic involved)

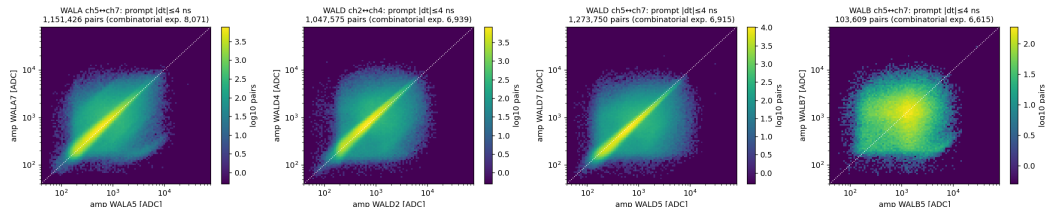
run224460: channel response to partner-tagged bar MIPs (top tagged by bottom MIP and vice versa — NO plastic involved; any pathology here is on the SiPM/analog side)



Each channel's response to bar-MIPs tagged by the *other end* of the same bars — plastic plays no role. Every channel shows a Landau, but: WALA uniformly low (24–29 mV tagged modes vs 35–43 mV on B); **WALD7 = 23 mV, WALD8 = 29 mV vs 38–48 mV for WALD g1–g3**. The pathology lives on the **SiPM/analog** side; the plastics are exonerated.

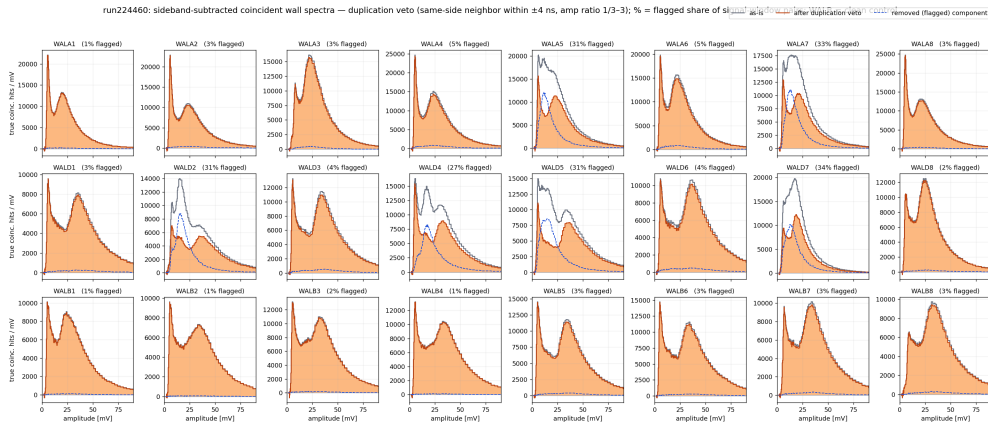
Odd-channel autopsy 2: it is signal *duplication*, not cross-talk

run224460: same-side cross-group prompt pairs — amplitude correlation (clean channels show only combinatorial haze, cross-talking ones a correlated band)



Prompt ($|dt| \leq 4$ ns) same-side cross-group pairs. WALA 5↔7, WALD 2↔4, WALD 5↔7: 1.0–1.3M pairs on a razor-sharp **equal-amplitude diagonal** (combinatorial expectation $\sim 7k$; mean ratio 0.8–1.1) — the *same pulse* appears on both channels. Clean reference (WALB 5↔7, right): combinatorial haze only. Capacitive cross-talk would sit far below the diagonal $\Rightarrow$ look for a shorted/split analog path (SiPM summing board, patch panel, or splitter) on those channel pairs.

The duplication explains the deformed spectra — and a veto cleans them



Veto: same-side neighbor hit within ± 4 ns with amplitude ratio 1/3–3. Flagged share of signal-window pairs:

27–34% on exactly the duplicated channels, 1–5% elsewhere, $\leq 3\%$ on all of WALB (control, spectra untouched). The removed component (blue dashed) is the deformation; survivors are clean Landaus (e.g. WALD7: broad hump $\rightarrow$ well-formed MIP at ~ 21 mV — its gain is still genuinely low). Channels keep 62–71% of their excess $\Rightarrow$ software cleaning is viable until the analog fix.

Channel verdicts & requests

Verdicts (all SiPM/analog side):

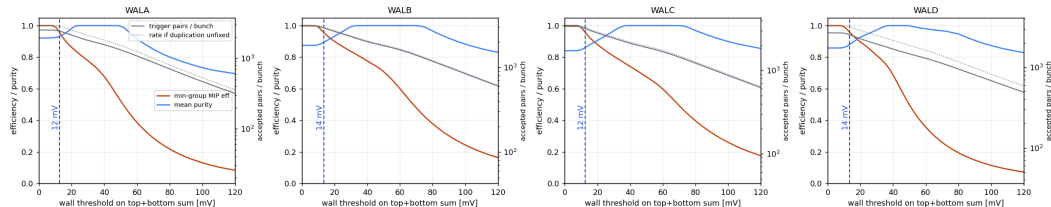
- ▶ WALA (all 8): gain $\sim 30\%$ low; several same-side channel pairs weakly correlated — check wall-A analog section / bias.
- ▶ WALA $5 \leftrightarrow 7$, WALD $2 \leftrightarrow 4$, WALD $5 \leftrightarrow 7$: equal-amplitude signal duplication (present and identical in 224404 and 224460) — inspect cabling/summing for these pairs.
- ▶ WALD ch7/ch8 (group 4): genuinely weak response (tagged MIP $\sim 40\%$ low) — bias/SiPM check.
- ▶ Plastics: healthy on all arms at the new operating point; PSSC1 (junk-dominated) and PSSD2 (low mode) worth an eye during the HV scan analysis.

Requests:

1. Inspect/swap analog paths of the three duplicated channel pairs before the next long run; a bench pulser check pinpoints the split.
2. HV/gain trim: wall A (all), wall D group 4; then re-derive MIP constants.
3. DAQ: why is the index tree empty in 224460?
4. Plastic HV scan analysis (in progress) → gain equalization; keep the 224404 operating point documented as the “do not use” reference.

Trigger thresholds: top+bottom sum, one threshold per wall

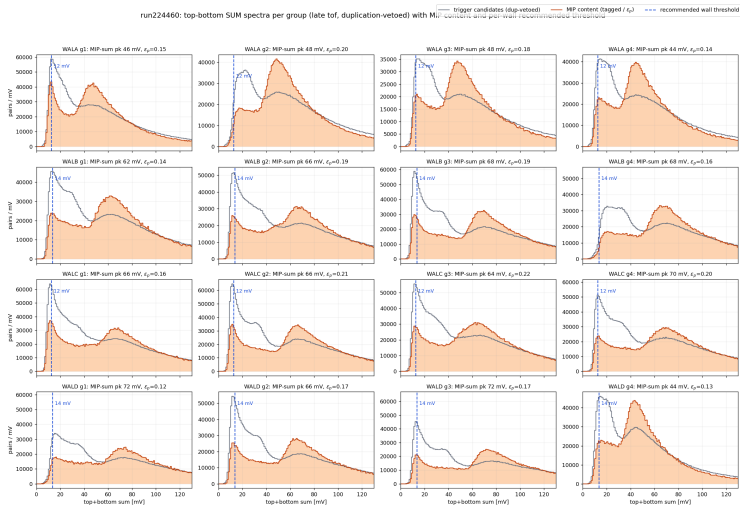
run224460: single-threshold-per-wall scan (dashed: recommended = weakest group at 95% eff)



wall	rec. thr	eff g1-g4	purity g1-g4	pairs/bunch	dup. unfixed
WALA	12 mV	.96/.99/.98/.98	.92-.94	2454	2878 (+17%)
WALB	14 mV	.97/.96/.96/1.0	.88-.91	2916	2981 (+2%)
WALC	12 mV	.96/.96/.97/.98	.84-.88	3184	3293 (+3%)
WALD	14 mV	.99/.96/.97/.97	.85-.94	2526	3168 (+25%)

Rule: weakest group of each wall kept at $\geq 95\%$ MIP efficiency; thresholds $\approx 20-30\%$ of the lowest group's MIP-sum peak (WALA low gain, WALD g4 weak). Duplication cannot be vetoed in hardware $\Rightarrow$ fix the analog paths before freezing rates. Details: `TRIGGER_THRESHOLDS.md`.

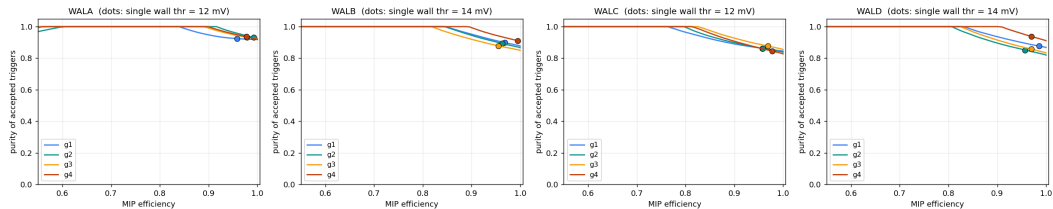
Trigger-candidate top+bottom sum spectra with recommended cut



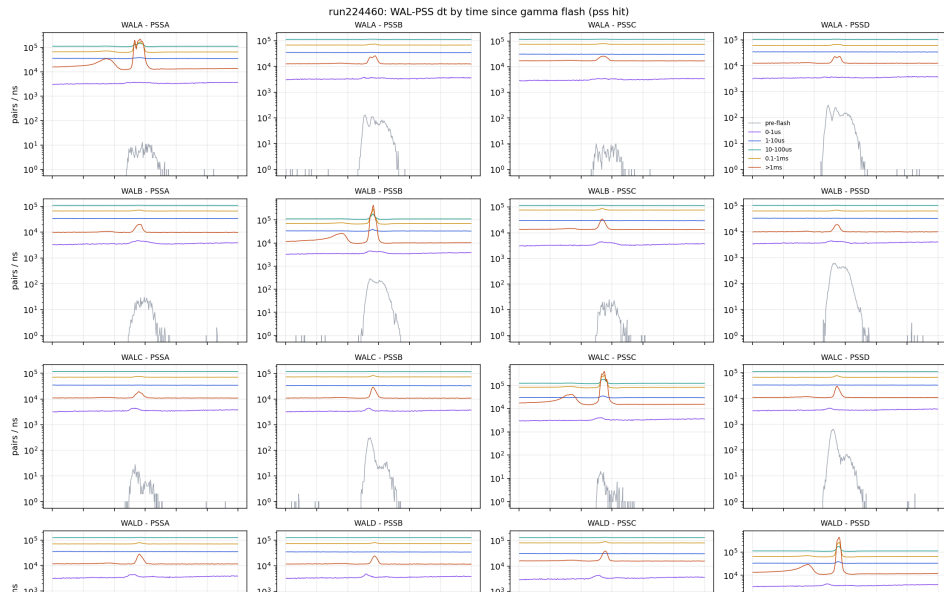
Log-scale variant: 18_trigger/trigsum_spectra.png. MIP content > candidates near the bump on some panels: ϵ_p grows with sum, so quoted purities at threshold are conservative.

Backup: purity vs efficiency per group

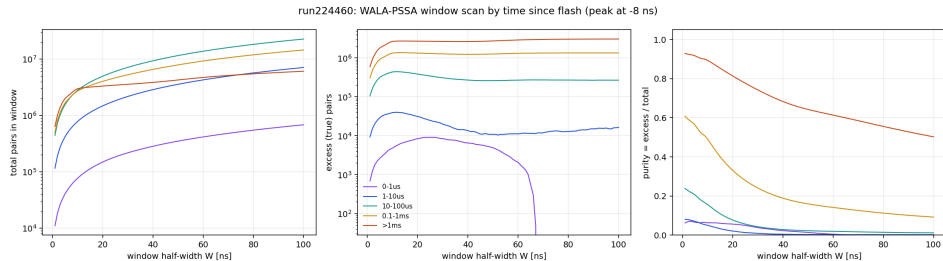
run224460: purity vs efficiency per group as the sum threshold varies



Backup: dt by time-since-flash region

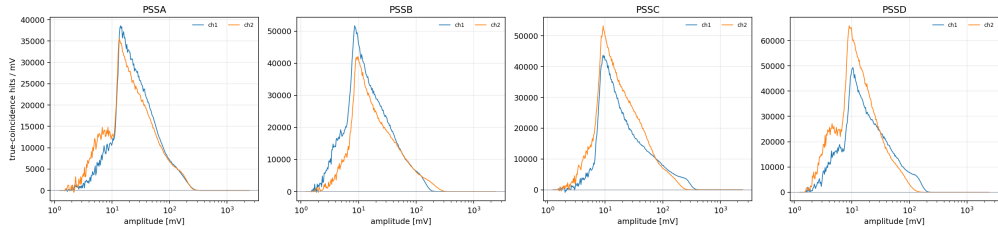


Backup: window scan & purity (arm A, now healthy)



Backup: plastic coincident spectra

run224460: sideband-subtracted coincidence amplitude spectra (plastic bars, tof > 0.1 ms, log axes)



Backup: rate stability over the run

